

SEATTLE KING COUNTY INTL, WA, USA

WMO: 727935

Lat: 47.530N

Lon: 122.301W

Elev: 6

StdP: 101.26

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24234

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.9	-1.2	-10.6	1.5	0.5	-7.3	2.0	1.7	10.0	9.5	8.7	8.7	1.7	140	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	29.9	18.4	27.9	17.7	26.2	17.1	19.3	28.1	18.5	26.3	17.7	24.7	3.3	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.3	11.6	20.8	15.4	10.9	20.6	14.7	10.4	20.3	54.6	28.2	51.9	26.2	49.6	24.5	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.1	5.9	-5.9	33.9	2.3	2.2	-7.5	35.4	-8.8	36.7	-10.1	37.9	-11.7	39.5		
(5)			-6.8	20.7	2.3	1.3	-8.5	21.7	-9.8	22.4	-11.1	23.1	-12.8	24.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	6.1	6.6	8.5	10.9	14.3	17.0	19.9	19.9	17.1	12.2	8.2	5.6	(6)
(7)		DBStd	5.86	3.21	2.89	2.75	2.71	3.04	2.92	2.76	2.36	2.54	2.54	3.14	3.15	(7)
(8)		HDD10.0	522	124	98	62	19	2	0	0	0	0	9	71	138	(8)
(9)		HDD18.3	2388	378	328	303	223	129	60	13	10	52	192	306	394	(9)
(10)		CDD10.0	1339	4	3	17	47	136	209	308	307	214	76	16	3	(10)
(11)		CDD18.3	163	0	0	0	1	6	19	62	59	17	0	0	0	(11)
(12)		CDH23.3	1234	0	0	1	8	57	172	495	397	104	1	0	0	(12)
(13)		CDH26.7	322	0	0	0	1	8	48	151	97	17	0	0	0	(13)
(14)	Wind	WSAvg	2.6	2.6	2.6	2.9	2.8	2.6	2.4	2.3	2.2	2.4	2.6	2.6	(14)	
(15)	Precipitation	PrecAvg	1015	145	98	109	71	51	42	17	23	43	105	152	165	(15)
(16)		PrecMax	1328	296	244	211	143	100	77	44	76	129	212	356	298	(16)
(17)		PrecMin	756	72	9	35	10	12	8	0	0	5	21	57	67	(17)
(18)		PrecStd	161	58	51	46	38	25	21	12	22	31	52	66	67	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	15.4	20.2	24.3	27.7	31.0	33.3	32.0	28.7	22.1	17.3	14.1	(19)
(20)			MCWB	11.1	9.5	11.5	14.5	16.5	18.6	19.7	18.9	17.5	14.7	13.4	11.6	(20)
(21)		2%	DB	12.9	13.5	16.9	20.2	24.8	27.8	30.5	29.1	26.2	19.3	15.1	12.4	(21)
(22)			MCWB	10.4	9.5	10.7	12.5	15.4	17.2	18.7	18.5	17.1	13.9	12.5	9.9	(22)
(23)		5%	DB	11.8	12.3	14.6	17.8	22.3	25.0	27.9	27.4	24.0	17.7	13.8	11.1	(23)
(24)			MCWB	9.4	9.0	9.9	11.4	14.6	16.0	17.9	17.9	16.5	13.4	11.6	8.9	(24)
(25)		10%	DB	10.6	11.0	12.9	15.9	20.0	22.7	26.1	25.6	22.3	16.3	12.6	10.0	(25)
(26)			MCWB	8.6	8.3	9.2	10.5	13.6	15.3	17.2	17.1	15.7	12.8	10.5	8.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	12.1	13.0	15.2	17.9	19.3	20.9	20.2	19.5	16.7	14.7	12.5	(27)
(28)			MCDWB	14.3	13.7	16.6	21.7	24.0	29.2	30.9	29.3	26.3	19.7	16.2	13.9	(28)
(29)		2%	WB	11.1	10.7	11.8	13.4	16.3	17.7	19.4	19.1	18.0	15.2	13.2	10.5	(29)
(30)			MCDWB	12.2	12.5	15.0	18.6	23.1	25.8	28.8	27.5	23.8	17.9	14.8	11.7	(30)
(31)		5%	WB	9.9	9.5	10.7	12.2	15.1	16.7	18.5	18.3	17.1	14.1	12.0	9.3	(31)
(32)			MCDWB	11.0	11.4	13.8	16.5	21.4	23.8	27.0	25.8	22.4	16.6	13.3	10.6	(32)
(33)		10%	WB	8.8	8.5	9.8	11.1	14.0	15.7	17.5	17.6	16.3	13.2	10.8	8.1	(33)
(34)			MCDWB	10.2	10.5	12.7	14.8	19.2	21.9	24.8	24.5	21.2	15.8	12.3	9.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.5	6.5	7.3	8.0	8.5	8.7	10.0	9.8	8.7	7.0	6.1	5.3	(35)
(36)			MCDBR	6.5	8.5	9.8	11.9	12.8	12.9	13.4	12.9	12.0	9.2	6.7	5.9	(36)
(37)			MCWBR	4.4	5.1	5.1	5.6	5.6	5.2	4.9	4.7	4.8	4.8	4.6	4.3	(37)
(38)		5% WB	MCDWB	5.0	6.2	8.1	10.1	11.2	11.5	12.4	11.6	10.1	7.0	5.6	5.2	(38)
(39)			MCWBR	4.1	4.4	4.9	5.2	5.4	5.1	5.0	4.6	4.6	4.5	4.3	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.319	0.327	0.360	0.378	0.368	0.354	0.360	0.348	0.332	0.323	0.333	(40)	
(41)		taud	2.486	2.545	2.540	2.419	2.387	2.441	2.499	2.490	2.515	2.563	2.537	2.402	(41)	
(42)		Ebn at Noon	754	855	900	894	886	895	904	884	864	826	757	687	(42)	
(43)		Edh at Noon	65	75	87	107	115	110	102	99	88	73	61	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.02	1.89	2.85	4.18	5.27	5.79	6.47	5.52	4.01	2.27	1.19	0.82	(44)	
(45)		RadStd	0.14	0.25	0.32	0.39	0.43	0.57	0.48	0.36	0.36	0.25	0.16	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page